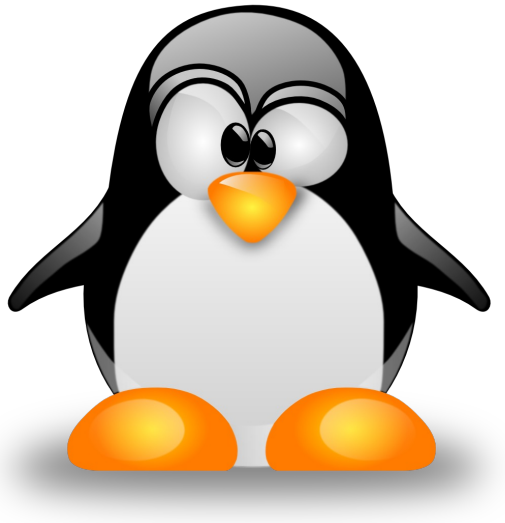


Mahidol University
Faculty of Information
and Communication Technology

ITCS254: Introduction to Linux

Ittipon (Art) Rassameeroj



Course Objectives

- Have a good working knowledge of Linux.
- Understand basic operating system operations.
- Navigate through major Linux distributions.
- Understand configurations and graphical interface of Linux.
- Use basic command-line operations.
- Understand and use shells.
- Explain common applications of Linux.

Instructor and LA

- Instructor: Ittipon Rassameeroj (Art)
  - https://ittiponr.github.io
  - My expertise
    - Computer systems & architecture
    - Cyber security
    - Operating/distributed systems, parallel/high-performance/cloud computing
  - Q & A and discussion: prefer to chat on MS Teams
- Lab assistant (LA): Kittikhun Thongkanchorn (P'Tae)

Course Outline: 15 Lectures 30 Hours

1. Linux philosophy & concepts and graphical interface
2. System configuration from GUI and common applications
3. Linux basic and system startup
4. Command line operations
5. Finding Linux documentation
6. Process
7. File operations
8. Text editors
9. User environments
10. Manipulating text
11. Network operations
12. Shell script (part 1)
13. Shell script (part 2)
14. Security
15. Basic system administrator

Grading

Midterm Exam **30%**

Final Exam **30%**

Labs **30%**

- We will have in-class lab assignments, and you must submit your work in every class.
- You can skip two labs for the whole class. If you do not submit more than that, your scores will be deducted.
- Late submission won't be accepted!

Self-Study for The Online Class (Linux Foundation Course) **10%**

- Please use your full English name to register for the course. We also use this course for our class materials.
- You must submit the course completion certificate on MyCourses by November 30, 2026.

Policies & References

- Attendance
  - We check attendance, but your attendance records do not affect eligibility to take the exams.
  - No need to submit leave forms.
- AI Policy
  - AI is not allowed during the class and exam!
  - You can use AI for self-study only.
- Class material references
  - https://training.linuxfoundation.org/training/introduction-to-linux/
  - “Linux Bible” by Christopher Hegus, 11th Edition, Wiley, 2025

Agenda

- What is Linux?
- Why Linux?
- Linux History
- Linux Philosophy and Community
- Terminology
- Linux Distribution
- Installation

What is Linux?

- Linux is an open source operating system that’s used in many areas of computing, initially developed on and for Intel x86-based personal computers.
- Linux is a Unix-like OS that's used in computers, servers, mobile devices, and more.
- It's known for being free, robust, secure, reliable, and versatile.
- It has been subsequently ported to an astoundingly long list of other hardware platforms, from tiny embedded appliances to the world's largest supercomputers.



Operating System Market Share and Facts

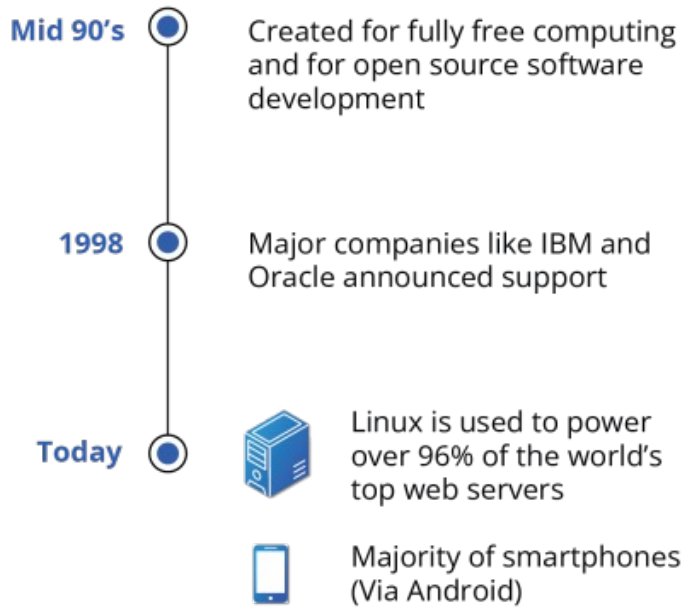
For Desktops

- Windows: 72.52%
- macOS: 14.68%
- Linux: 4.05%
- Unknown: 6.47%

For Servers

- Linux: 77 - 80%
- Microsoft Windows: 20%
- Others: 2 - 3%

- The majority of smartphones (via the Android system, which is built on top of Linux)
- More than 90% of the public cloud workload, and all of the world's most powerful supercomputers.
- Linux is used on all of the world's 500 fastest Supercomputers.
- Linux also leads other big iron systems, such as Mainframe.
- About 32.8 million Linux users worldwide.



How Linux Differs from Others?

- You cannot see the code used to create the OS, and therefore, you cannot change the OS at its most basic levels if it doesn't suit your needs, and you can't use the OS to build your own OS from source code.
- You cannot check the code to find bugs, explore security vulnerabilities, or simply learn what that code is doing.
- You may not be able to plug your own software easily into the OS if the creators of that system don't want to expose the programming interfaces you need to the outside world.

Linux World

- Mobile OS: Android
- Network device OSs
  - Cisco IOS XR, Arista EOS, Juniper Junos, Cumulus Linux, etc.
- Cloud and virtualization platforms
  - Docker and Kubernetes hosts
  - AWS, Azure, Google Cloud, etc.
- Consumer electronics and IoT
  - Smart TVs (e.g., Android TV and LG's webOS) and Set-Top Boxes
- Desktop systems
  - Chrome OS, SteamOS
- Automotive systems
  - Automotive Grade Linux (AGL)
  - Tesla Autopilot and Interface Systems
- Supercomputers, HPC for scientific computing
- Personal fitness devices, such as FitBit
- Medical devices

History [1/2]

Early Inspiration

- 1969: Unix, the precursor to Linux, was developed at Bell Labs by Ken Thompson and Dennis Ritchie.
- 1983: Richard Stallman launched the GNU Project to create a free Unix-like operating system.

Creation of Linux

- 1991: Linus Torvalds, a student at the University of Helsinki, created the Linux kernel as a free and open-source alternative to proprietary Unix systems.
- October 1991: Linux 0.02 was publicly released, with a functional kernel but limited features.

Rapid Growth

- 1992: Linux was re-licensed using the General Public License (GPL) by GNU (a project of the Free Software Foundation or FSF, which promotes freely available software), allowing for collaborative development.
- 1994: Linux 1.0 was released, marking the first stable kernel version.



Linus Torvalds

History [2/2]

Enterprise Adoption

- Late 1990s: Companies like Red Hat and SUSE (pronounced Su-Sa) began creating enterprise-grade Linux distributions.
- 2000s: Linux gained traction in servers, supercomputers, and embedded systems.

Modern Era

- 2008: Android, based on the Linux kernel, launched, revolutionizing the mobile OS market.
- Present: Linux powers cloud computing, IoT, supercomputers (over 90% of the top 500), and enterprise systems.

UNIX-Like Operating System Examples



Linux Philosophy

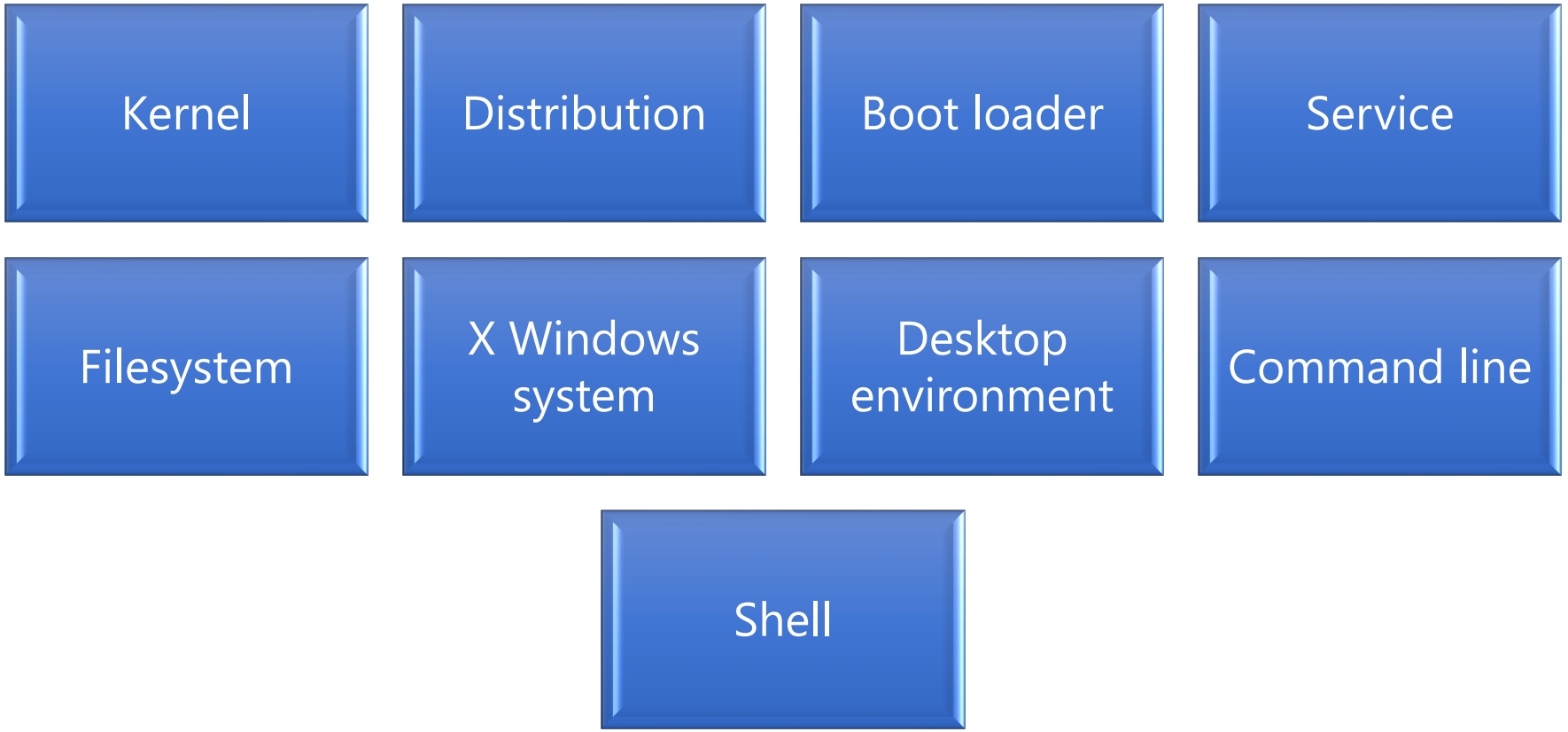
- Linux is constantly enhanced and maintained by a network of developers from all over the world collaborating over the Internet, with Linus Torvalds at the head.
- Technical skills, a desire to contribute, and the ability to collaborate with others are the only qualifications for participating.
- Linux borrows heavily from the well-established family of UNIX operating systems.
- Linux was inspired by UNIX, but it is not UNIX.
- It was written to be a free and open-source alternative; at the time, UNIX was designed for computers much more powerful than PCs, and furthermore, it was quite expensive.
- Whenever possible, Linux makes its components available via files or objects that look like files.
    - Processes, devices, and network sockets are all represented by file-like objects and can often be worked with using the same utilities used for regular files.
- Linux is a fully multitasking, multiuser OS with built-in networking and service processes known as daemons in the UNIX world.

Linux Community

- Over 12,000 developers from over 1,200 companies.
- It is the single largest software project.
- About 10 patches a day.
- Every 3 months, another version of Linux is released.



Linux Terminology [1/5]



Linux Terminology [2/5]

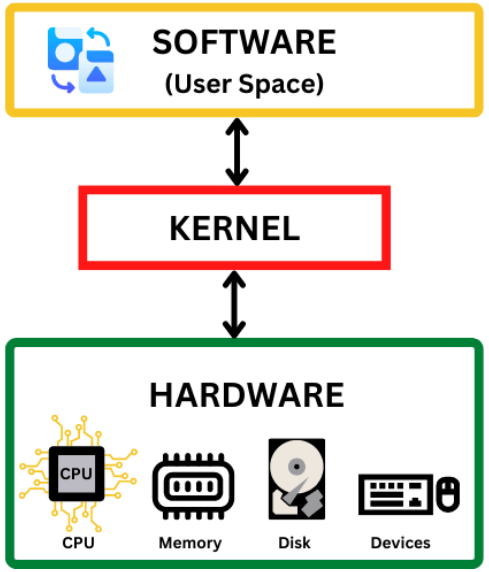
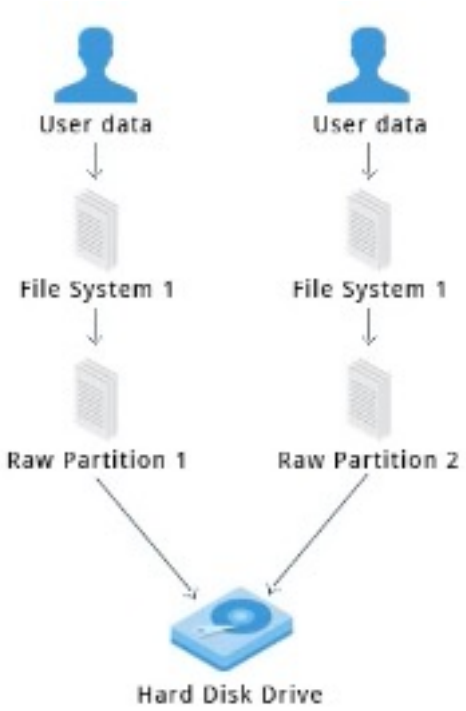


Image source: https://linuxsimply.com/what-is-kernel/

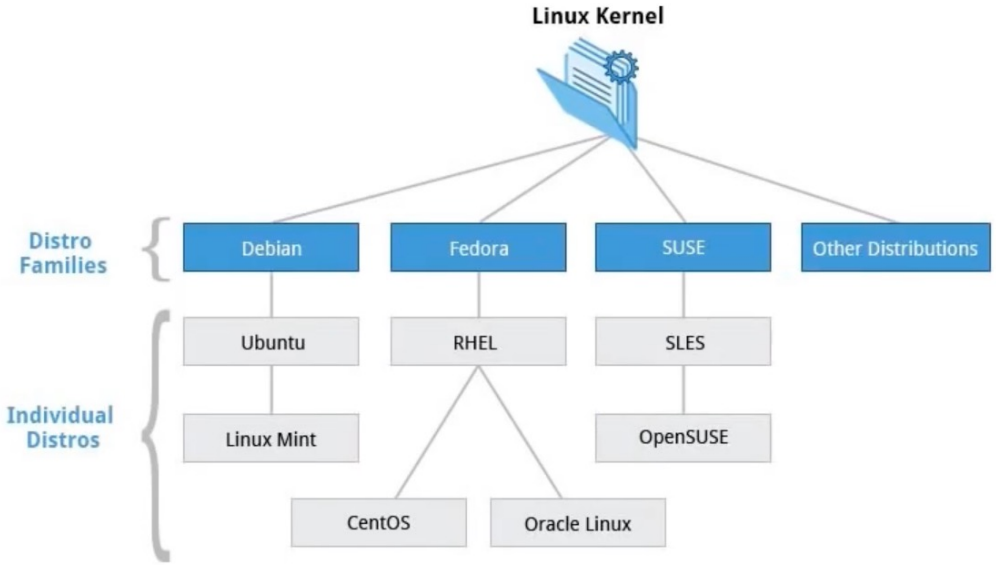
Filesystem is a method for storing and organizing files,

such as, ext3, ext4, FAT, XFS, NTFS, and Btrfs.

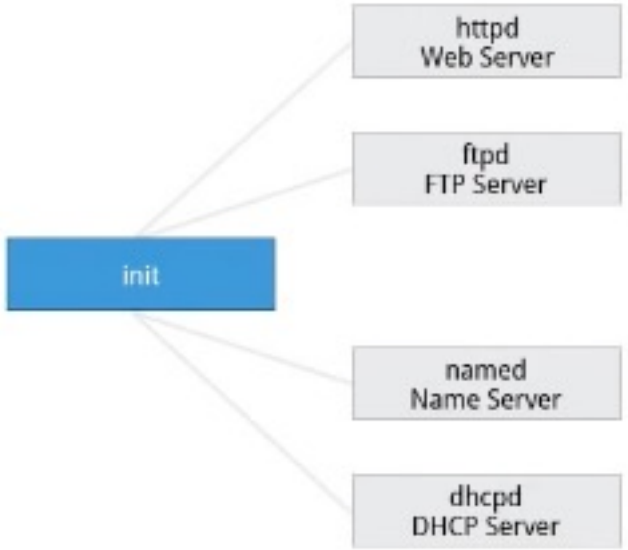


Linux Terminology [3/5]

- Distribution is a collection of software making up a Linux-based OS.
- Boot loader is program that boots the OS, such as GRUB and ISOLINUX.
- Service is program that runs as a background process.



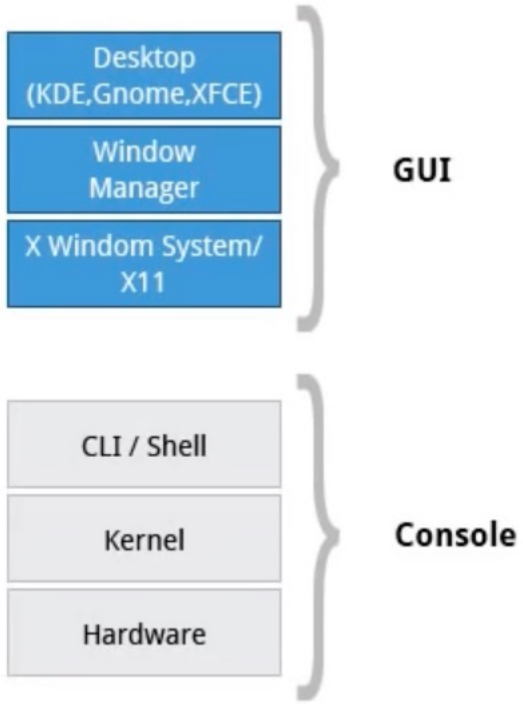


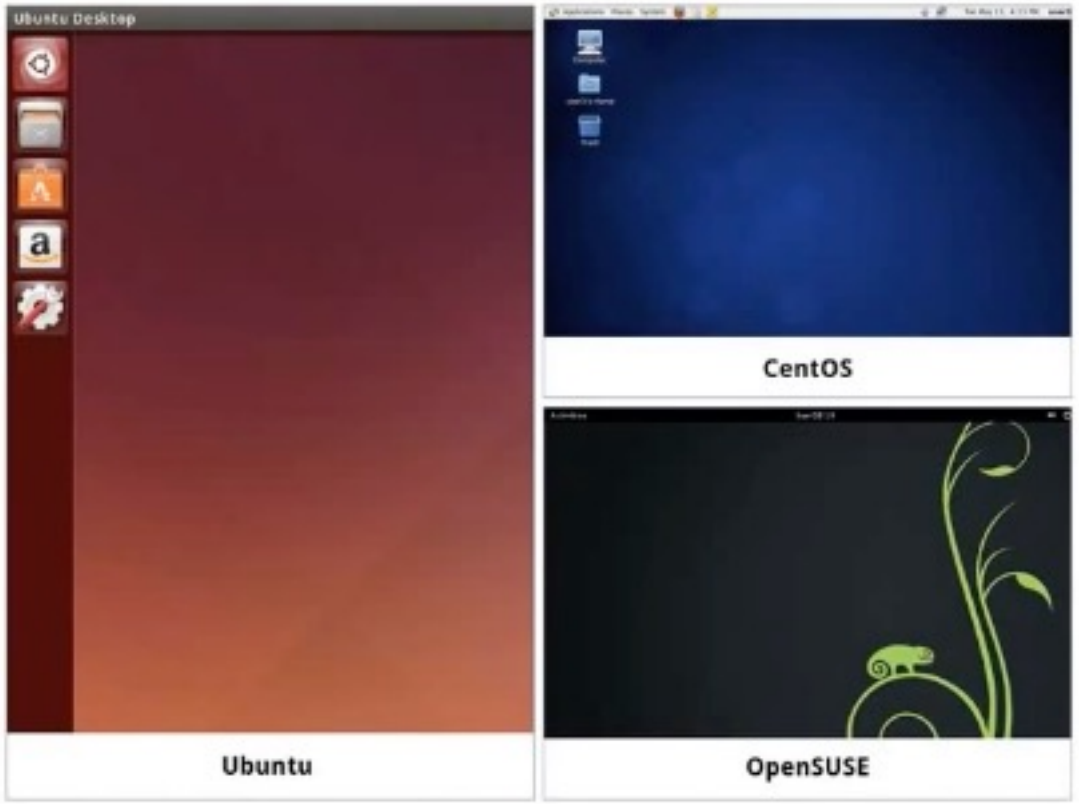


Linux Terminology [4/5]

X Window System: provides the standard toolkit and protocol to build a graphical subsystem on nearly all Linux systems.

Desktop environment: graphical user interface on top of the OS, such as GNOME, KDE, Xfce, and Fluxbox





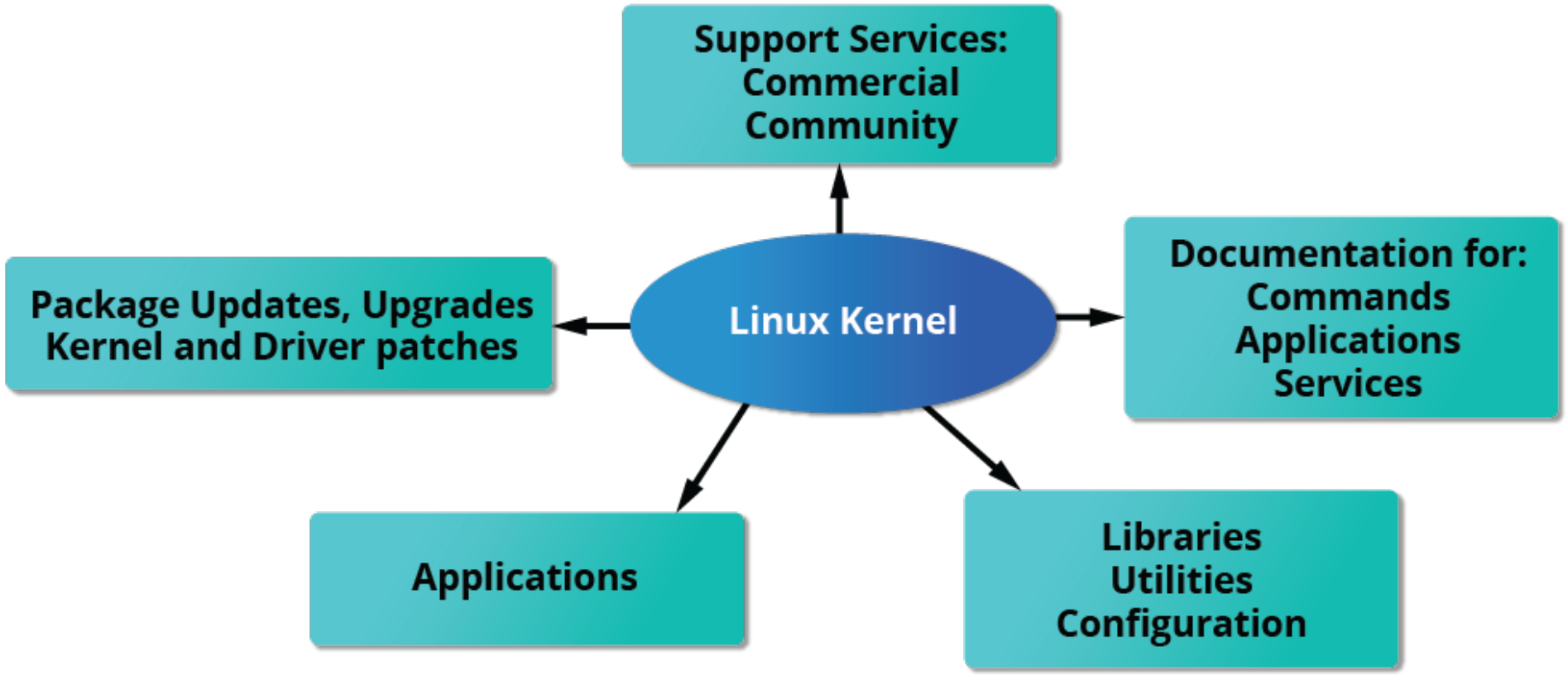
Linux Terminology [5/5]

- Command line is interface for typing commands on top of the OS.
- Shell is command line interpreter that interprets the command line input and instructs the OS to perform any necessary tasks and commands.

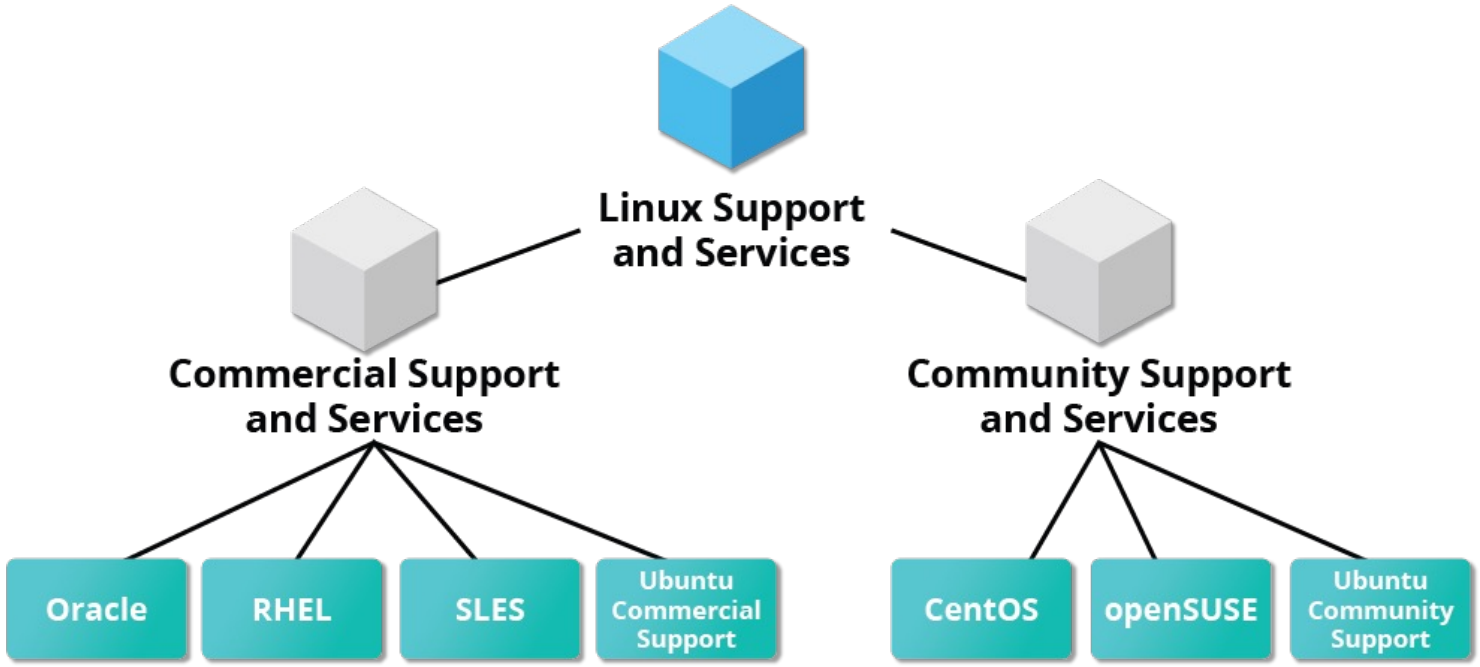
Linux Distribution (Distro)

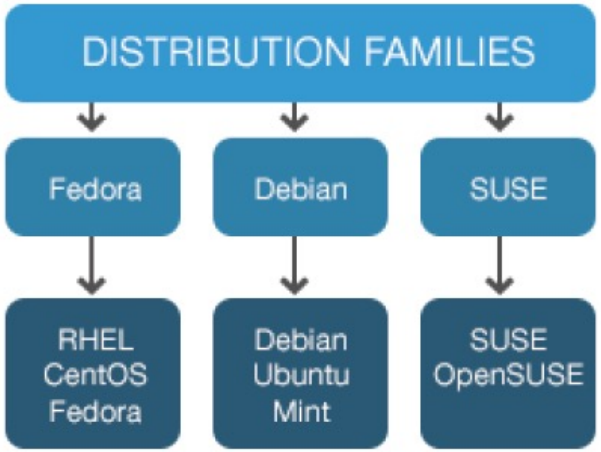
- A full Linux distribution consists of the kernel plus a number of other software tools for file-related operations, user management, and software package management.
- Each of these tools provides a part of the complete system.
  - Each tool is often its own separate project, with its own developers working to perfect that piece of the system.
- While the most recent Linux kernel (and earlier versions) can always be found in the Linux Kernel Archives, Linux distributions may be based on different kernel versions.
- Examples of other essential tools and ingredients provided by distributions include the C/C++ and Clang compilers, the gdb debugger, the core system libraries applications need to link with in order to run, the low-level interface for drawing graphics on the screen, as well as the higher-level desktop environment, and the system for installing and updating the various components, including the kernel itself.

Distribution Roles



Linux Distribution



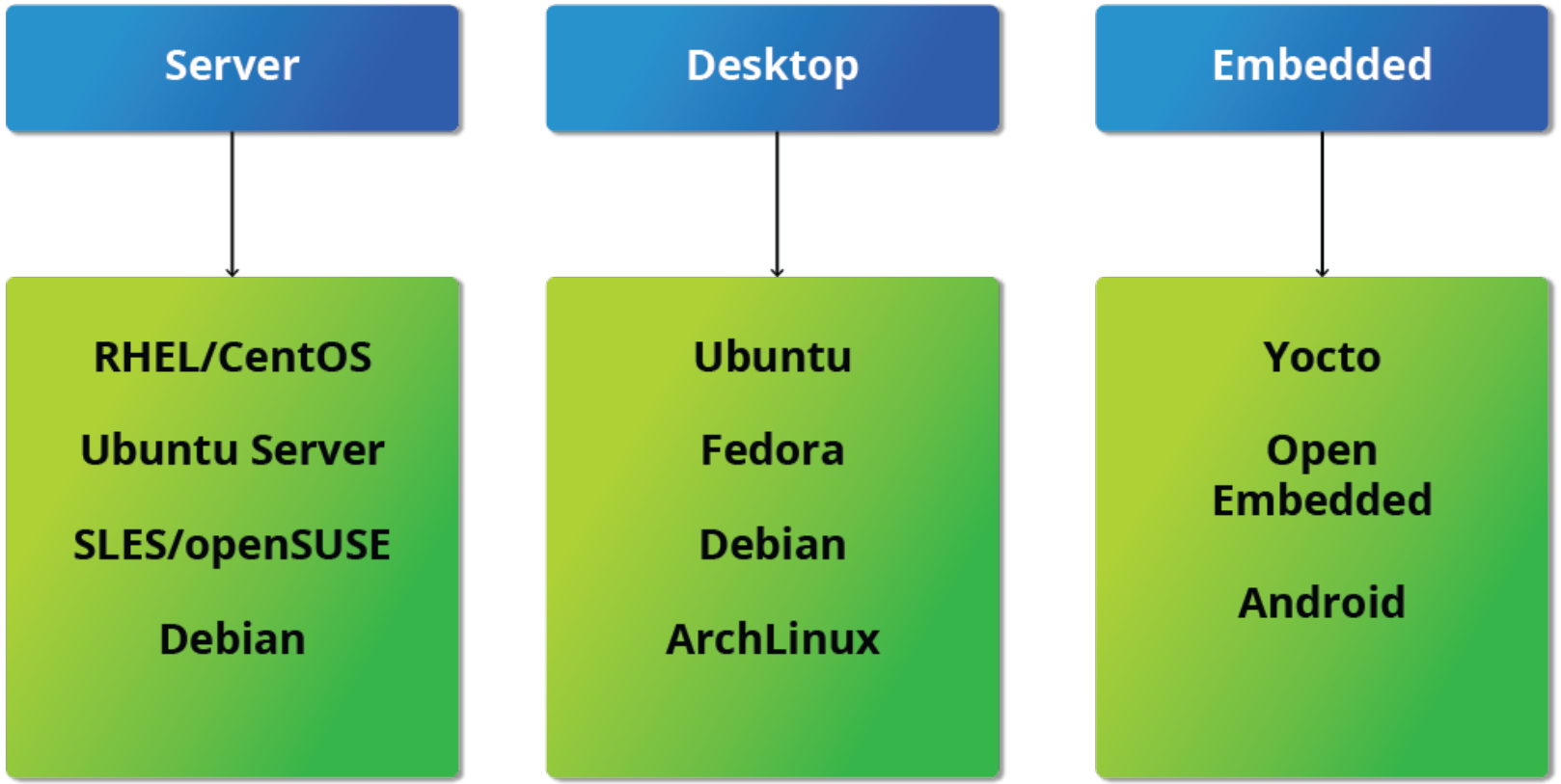


RHEL = Red Hat Enterprise Linux
SLES = SUSE Linux Enterprise Server

Linux Distribution

- Note that new versions of CentOS disappeared at the end of 2021 in favor of CentOS Stream. However, at least two new RHEL-derived substitutes, AlmaLinux and Rocky Linux, have established a healthy foothold.
- The RHEL variants, such as CentOS and AlmaLinux, are designed to be binary-compatible with RHEL; i.e., in most cases, binary software packages will install properly across the distributions.
- Ubuntu and Fedora are widely used by developers and are also popular in the educational realm.
- Many commercial distributors, including Red Hat, Ubuntu, SUSE, and Oracle, provide long-term fee-based support for their distributions, as well as hardware and software certification.
- All major distributors provide update services for keeping your system primed with the latest security and bug fixes and performance enhancements, as well as provide online support resources.
- https://en.wikipedia.org/wiki/Linux_distribution

Choosing a Linux Distribution

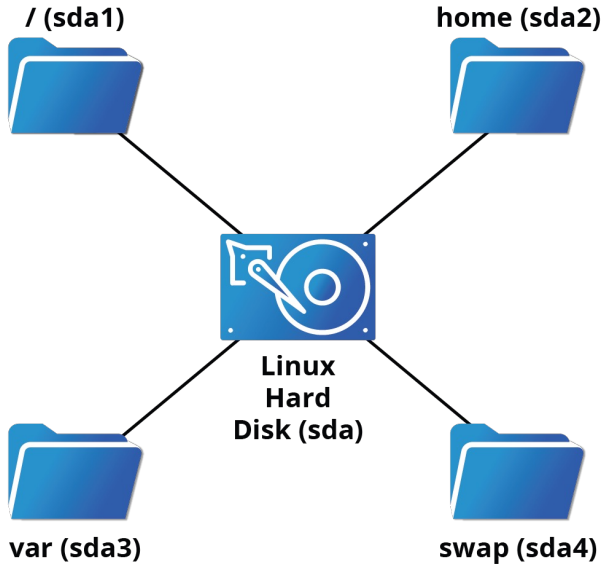


Questions to Ask When Choosing a Distribution

- What is the main function of the system (server or desktop)?
- What types of packages are important to the organization? For example, web server, word processing, etc.
- How much storage space is required, and how much is available? For example, when installing Linux on an embedded device, space is usually constrained.
- How often are packages updated?
- How long is the support cycle for each release? For example, LTS releases have long-term support.
- Do you need kernel customization from the vendor or a third party?
- What hardware are you running on? For example, it might be X86, RISC-V, ARM, PPC, etc.
- Do you need long-term stability? Or can you accept (or need) a more volatile cutting-edge system running the latest software versions?

Installation Steps [1/3]

- Planning
- Software choices
  - Distributions provide options for adding categories of software.
  - Common applications (such as the Firefox web browser and LibreOffice office suite), developer tools (like the vi and emacs text editors), and other popular services (such as the Apache web server tools or MySQL database) are usually included.
  - For any system with a graphical desktop, a chosen desktop (such as GNOME or KDE) is installed by default.
  - All installers set up some initial security features on the new system.
    - One basic step consists of setting the password for the superuser (root) and setting up an initial user.
    - Some distributions will also install more advanced security frameworks, such as SELinux or AppArmor.



Installation Steps [2/3]

- Installation source
  - Linux distributions are provided on removable media, such as USB drives and CDs or DVDs.
  - Most distributions also support booting a small image and downloading the rest of the system over the network.
    - These small images are usable on media, or as network boot images, in which case it is possible to perform an install without using any local media.
  - Many installers can do an installation completely automatically, using a configuration file to specify installation options.
    - This file is called a Kickstart file for Red Hat-based systems, an AutoYAST profile for SUSE-based systems, and a Preseed file for Debian-based systems.

Installation Steps [3/3]

- Installation process
  - The actual installation process is pretty similar for all distributions.
  - After booting from the installation media, the installer starts and asks questions about how the system should be set up. These questions are skipped if an automatic installation file is provided. Then, the installation is performed.
  - Finally, the computer reboots into the newly-installed system, and additional configuration questions are asked.
  - Most installers have the option of downloading and installing updates as part of the installation process; this requires Internet access.

Lab 1: Install Ubuntu Desktop on VirtualBox

- Download Ubuntu 26.04 LTS or higher from **only one of these options**:
  - https://drive.ict.mahidol.ac.th/index.php/s/R41W5vxdawQE5jk
  - https://ubuntu.com/download/desktop
- How to install Ubuntu on VirtualBox
  - https://selftuts.in/install-ubuntu-24-on-virtualbox/
  - https://techarry.com/how-to-install-ubuntu-26-04-lts-on-virtualbox/
- Official manual of Ubuntu installation: https://ubuntu.com/tutorials/install-ubuntu-desktop
- Settings
  - Virtual machine name (in VirtualBox settings): ITCS254
  - Name: itcs254
  - Username: itcs254
  - Password: itcs254-linux